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18 July 1980
Superseding
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14 December 1973

# MILITARY SPECIFICATION

## MOLDING COMPOUND, PHENOLIC RESIN, ASBESTOS REINFORCED

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of asbestos reinforced phenolic resin molding compound used as insulation in rocket motors.

## 2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-1188 | Commercial Packaging of Supplies and Equipment |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-ED, Redstone Arsenal, AL 35809, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9330

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functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS

| | |
|---|---|
| ASTM D 299 | Asbestos Yarns, Specification for |
| ASTM D 647 | Design of Molds for Test Specimens of Plastic Molding Materials, Recommended Practice for |
| ASTM D 651 | Tensile Strength of Molded Electrical Insulating Materials, Method of Test for |
| ASTM D 792-66 | Specific Gravity and Density of Plastics by Displacement, Test Methods for |
| ASTM D 796 | Compression Molding Test Specimens of Phenolic Molding Compounds, Recommended Practice for |
| ASTM D 1895-69 | Apparent Density, Bulk Factor, and Pourability of Plastic Materials, Test Methods for |

(Application for copies should be addrssed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.



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3. REQUIREMENTS.

3.1 Material. The material shall be a heat-resistant, mineral-filled asbestos reinforced phenolic resin molding compound.

3.1.1 Phenolic resin. The phenolic resin shall be a two-step high temperature resin containing a siliceous type mineral filler. The mineral filler may range between plus four and minus two percent by weight.

3.1.2 Reinforcement. The reinforcement shall be plain asbestos yarn, Grade AAAA, in accordance with ASTM D 299.

3.1.3 Size. Unless otherwise specified (see 6.2), the material shall be furnished in 5/8 inch length chopped strands.

3.2 Physical properties. The physical properties of the molding compound shall be in accordance with Table I.

Table I

Physical Properties

| Property | Minimum | Maximum | Test Method |
|---|---|---|---|
| Volatiles content, percent by weight | 3 | 7 | 4.3.2 |
| Resin solids, percent by weight | 44 | 48 | 4.3.2 |
| Tensile strength, (of molded produce) pounds per square inch | 9,000 | -- | 4.3.3 |
| Specific gravity | 1.70 | 1.74 | 4.3.4 |
| Bulk factor | 5 | 8 | 4.3.5 |

3.3 Workmanship. The material shall be uniform in composition, size, and color. There shall be no foreign material or other defects that could adversely affect the suitability, use, or life of the material.



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## 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. All inspections specified herein are classified as quality conformance inspections.

4.2.1 Inspection lot. An inspection lot shall consist of molding compound manufactured by one process under essentially uniform conditions by one manufacturer and offered for inspection at any one time.

4.2.2 Sampling. Sampling for visual examination shall be 100 percent of the lot. Failure of any sample to meet the visual examination shall be cause for rejection of the sample. Sampling for all other examinations and tests shall be in accordance with MIL-STD-105, Level II, single sampling, AQL 1.0 percent. Sufficient material shall be taken from a random location in each container selected.

4.2.3 Procedure. Samples taken in accordance with 4.2.2 shall be subjected to the examinations and tests as listed in Table II.

Table II

Examinations and Tests

| Examination | Test Paragraph | Requirement Paragraph |
|---|---|---|
| Visual | 4.3.1 | 3.3, Section 5 |
| Votatile content | 4.3.2 | 3.2 |
| Resin content | 4.3.2 | 3.2 |



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| | | |
|---|---|---|
| Tensile strength | 4.3.3 | 3.2 |
| Specific gravity | 4.3.4 | 3.2 |
| Bulk factor | 4.3.5 | 3.2 |

4.3 Inspection methods.

4.3.1 Visual examination. Visually examine all containers of the lot to determine compliance with 3.3 and Section 5.

4.3.2 Volatile and resin solids content.

4.3.2.1 Apparatus.

a. Crucibles. The crucibles shall be fired to a temperature of 782 to 793 degrees Celsius (C) (1440 to 1460 degrees Fahrenheit (F)) and weighed to the nearest 0.001 gram.

b. Oven. The oven shall be a circulating air oven capable of maintaining a temperature of 160 to 166 degrees Celsius (C) (320 to 330 degrees Fahrenheit (F)).

c. Muffle furnaces. The muffle furnaces shall be capable of maintaining a temperature of 471 to 482 and 782 to 793 degrees Celsius (C) (880 to 900 and 1440 to 1460 degrees Fahrenheit (F)).

4.3.2.2 Test Procedure. Weigh 5 to 6 grams to the nearest 0.001 gram of the sample into a previously fired and tared crucible. Condition the specimen and crucible in the oven for not less than 20 minutes at 160 to 166 degrees Celsius (C) (320 to 330 degrees Fahrenheit (F)). Cool the specimen and crucible in the desiccator and determine the specimen weight.

4.3.2.3 Calculation. Volatile and resin solids content shall be calculated as the average content for three test specimens.

4.3.2.3.1 Volatile content. The percent volatile content shall be calculated as follows:

$$\text{Volatile content, percent} = \frac{(W_1 - W_2) \times 100}{W_1}$$

where: $W_1$ = Original specimen weight



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$W_2$ = Specimen weight after conditioning for 20 minutes at 160 to 166 degrees Celsius (C) (320 to 330 degrees Fahrenheit (F))

4.3.2.3.2 Resin solids content. The resin solids content shall be calculated as follows:

$$\text{Resin solids content, percent} = 100 - \frac{100 \times W_3}{0.86 \times W_2}$$

Where: $W_3$ = Specimen weight after conditioning for 20 minutes at 160 to 166 degrees Celsius (320 to 330 degrees Fahrenheit (F)), 4 hours at 471 to 482 degrees Celsius (C) (880 to 900 degrees Fahrenheit (F)), and 1 hour at 782 to 793 degrees Celsius (C) (1440 to 1460 degrees Fahrenheit (F)).

4.3.3 Tensile strength.

4.3.3.1 Preparation of test specimen. Molds for test specimens for determining tensile strength shall be in accordance with Figure 3 of ASTM D 647. Test specimens shall be prepared in accordance with ASTM D 796, Molding pressure shall be 2500 plus or minus 500 pounds per square inch (psi). Test specimens shall be cured for not less than 10 minutes at a temperature of 141 to 152 degrees Celsius (C) (285 to 305 degrees Fahrenheit (F)).

4.3.3.2 Test Method. Tensile strength shall be determined in accordance with ASTM D 651. Calculate tensile strength as the average strength for five test specimens.

4.3.4 Specific gravity.

4.3.4.1 Preparation of test specimens. Test specimens shall be prepared as specified in 4.3.3.1.

4.3.4.2 Test method. Specific gravity shall be determined in accordance with ASTM D 792. Calculate specific gravity as the average specific gravity for three test specimens.



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4.3.5 Bulk factor. Bulk factor shall be determined in accordance with ASTM D 1895. Calculate bulk factor as the average bulk factor for three test specimens.

* 5. PACKAGING

* 5.1 Preservation, packing, unitization and marking. Level A, B or Commercial packaging (see 6.2).

* 5.1.1 Level A or B. Preservation, packing, unitization, and marking shall be in accordance with packaging data sheet APN MIL-M-46891 and specifications references therein.

* 5.1.2 Commercial. Preservation, packing, unitization and marking shall be in accordance with MIL-STD-1188. Additional marking shall include shelf-life and storage limitations, if applicable, in accordance with MIL-STD-129.

* 5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not limited to, the following information:

a. Manufacturer's name and location.

b. Material trade name.

c. Net weight.

d. Lot number, batch number, and date of manufacture.

e. Number and revision letter of this specification.

f. Shelf life and storage limitations, if applicable.

6. NOTES

6.1 Intended use. One type of asbestos reinforced phenolic resin molding compound for use as insulation in rocket motors.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Minimum lot size, if applicable.

c. Place of delivery.

d. Size of container.



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e. Place of inspection.

f. Disposition of test data.

* 6.3 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corresions, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-M-46891A(MI)

MOLDING COMPOUND, PHENOLIC RESIN, ASBESTOS REINFORCED

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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